

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2025

CS2 - Risk Modelling and Survival Analysis
(Paper A)

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** A climate risk actuary is analysing the temperatures in the city of Jaisalmer, the hottest city in India. The current record of highest temperature in the city is a blistering 52 degrees Celsius observed in the year 2016. The actuary feels that that this record might be broken in the year 2025 and is intending to use Extreme Value Theory (EVT) to model this.

The actuary has obtained temperature data for Jaisalmer for the past 10 years. He has performed EVT using Generalised Pareto Distribution by selecting only those days where the temperature exceeded 50 degrees Celsius. This has resulted in the hottest 150 days being selected for the analysis.

Following parameters were obtained from the EVT analysis:

$$\begin{aligned}\beta \text{ (scale parameter)} &= 15, \\ \gamma \text{ (shape parameter)} &= 3.\end{aligned}$$

What is the probability that the current record of 52 degrees Celsius would be broken in the year 2025? You may assume that one year has 365 days.

- A. 0.35%
- B. 2.14%
- C. 3.61%
- D. 12.23%
- E. None of the above

[2]

- Q. 2)** Consider the time series process X_t given by:

$$X_t = aX_{t-1} + \frac{1}{2} X_{t-2} + e_t$$

where e_t is a sequence of independent and identically distributed $N(0, \sigma^2)$ random variables.

Find the value of 'a' for which the process X_t is I (1).

Note: Assume that both the roots of the characteristic polynomial are greater than zero.

- A. $a = 4$
- B. $a = 2$
- C. $a = 1$
- D. $a = \frac{1}{2}$
- E. $a = \frac{3}{4}$

[2]

- Q. 3)** A local authority responsible for maintaining marriage, separation and death records of its citizens, has classified its citizens into three states:

UM	: Un-married citizens (includes citizens who have taken divorce)
M	: Married citizens
D	: Dead citizens

Every year, the local body collects data relating to each citizen and denotes the state of the citizen.

Year-to-year changes in the classification are modelled using a Markov chain with the following 1-year transition probabilities:

	UM	M	D
UM	0.20	0.70	0.10
M	0.35	0.50	0.15
D	0	0	1

Based on the given transition matrix, what is the probability that an unmarried citizen (who is currently in “UM” state) will stay unmarried for the rest of his life?

- A. 0.125
- B. 0.20
- C. 0.20^∞
- D. 0.020
- E. 0.25

[2]

- Q. 4)** A life insurance company has conducted a mortality investigation for people from ages 60 years to 70 years. The raw data has been graduated by reference to a standard table of assured lives.

Following notations have been used:

A_i = Actual deaths at age i

E_i = Expected deaths at age i

$z_i = (A_i - E_i) / \sqrt{E_i}$

Following aggregates have been obtained:

$$\sum_{i=60}^{70} z_i = 0.349668,$$

$$z_{70} = 0.643078, z_{60} = -0.37165,$$

$$\sum_{i=60}^{70} z_i^2 = 4.62523,$$

$$\sum_{i=60}^{69} (z_i * z_{i+1}) = 2.322974.$$

What is the value of the test-statistic for the Serial Correlations Test of graduation at lag 1?

- A. 1.77
- B. 1.82
- C. 0.56
- D. 2.14
- E. 3.68

[3]

- Q. 5)**

A data scientist is using K-means clustering algorithm for initial division of data relating to FinTech companies. Five cluster centres have been identified based on Price Earnings Ratio and Return on Investment.

Consider a newly listed financial services company ABC. which has P/E ratio equal “c” times and ROI equal to “d” %. A cluster centre needs to be assigned to this company based on its distance from the five cluster centres.

Consider a cluster centre $A \equiv (e, f)$ and let ABC . be denoted by $B \equiv (c, d)$.

Consider the following statements:

1. Euclidean distance between points A and B will always be lower than the sum of absolute distance between points A and B.
2. Euclidean distance between points A and B will always be higher than the sum of absolute distance between points A and B.
3. Euclidean distance between points A and B will be exactly equal to the sum of absolute distance between points A and B if ABC has P/E ratio exactly equal to the P/E ratio of Cluster A.
4. Euclidean distance between points A and B will be exactly equal to the sum of absolute distance between points A and B if ABC has ROI exactly equal to the ROI of Cluster A.

Which of the given statement(s) is/are TRUE?

- A. Statement 1 only
- B. Statement 2 only
- C. Statement 3 only
- D. Statement 4 only
- E. Statements 3 and 4 only

[2]

Q. 6) A bi-variate copula from the Archimedean family has the following generator function:

$$\Psi(t) = \ln\left(\frac{1 - \theta(1-t)}{t}\right) \quad \dots\dots -1 \leq \theta \leq 1$$

Which of the following represents the correct definition of C [u, v] for this bi-variate copula?

- A. $C[u, v] = \exp\left[-\left((-\log(u))^\theta + (-\log(v))^\theta\right)^{1/\theta}\right]$
- B. $C[u, v] = -\frac{1}{\theta} \ln\left[1 + \frac{(\exp(-\theta u) - 1)(\exp(-\theta v) - 1)}{\exp(-\theta) - 1}\right]$
- C. $C[u, v] = \frac{uv}{1 - \theta(1-u)(1-v)}$
- D. $C[u, v] = [u^{-\theta} + v^{-\theta} - 1]^{-1/\theta}$
- E. $C[u, v] = 1 - [(1-u)^\theta + (1-v)^\theta - (1-u)^\theta * (1-v)^\theta]^{1/\theta}$

[3]

Q. 7) BROMA, a manufacturer of electronic items manufactures tri-pods for stabilizing, supporting and elevating cameras. Each tri-pod has three legs. It sells 1,000 tripods in a year. Each tripod has a binomial probability of “p” of being involved in a fall and on fall, there is a binomial probability of 0.1 for each leg being replaced at a cost of INR 500. Let S denote the total cost of replacement of tri-pod legs in a year.

Which of the following represents the moment generating function of S i.e. $M_S(t)$?

- A. $(1 - 0.729p + 0.243pe^{500t} + 0.027pe^{1000t} + 0.001pe^{1500t})^{1000}$
- B. $(1 + 0.729p + 0.243pe^{500t} + 0.027pe^{1000t} + 0.001pe^{1500t})^{1000}$
- C. $(1 + 0.271p + 0.243pe^{500t} + 0.027pe^{1000t} + 0.001pe^{1500t})^{1000}$
- D. $(1 - 0.271p + 0.243pe^{500t} + 0.027pe^{1000t} + 0.001pe^{1500t})^{1000}$
- E. None of the above

[3]

Q. 8) A moving average time series is defined by the relationship:

$$X_t = a + b e_t + c e_{t-1} + d e_{t-2}$$

In the given time series, e_t is a sequence of independent and identically distributed $N(0, \sigma^2)$ random variables and denotes white noise.

The autocorrelation function at lag 2 i.e. ρ_2 is given by which of the following expressions?

- A. $ab / (b^2 + c^2 + d^2)$
- B. $bc / (b^2 + c^2 + d^2)$
- C. $cd / (b^2 + c^2 + d^2)$
- D. $b^2 / (b^2 + c^2 + d^2)$
- E. $bd / (b^2 + c^2 + d^2)$

[2]

Q. 9) Y_t follows the below time series process:

$$Y_t^2 - \beta_1 e_t^2 Y_{t-1}^2 = 2(Y_t - \beta_1 e_t^2 Y_{t-1})\mu - (1 - \beta_1 e_t^2)\mu^2 + \beta_0 e_t^2$$

In the given time series, e_t are independent standard normal variables and they are also independent from $Y_t, Y_{t-1}, \dots$

Which of the following represents the correct value of $E(Y_t)$ is –

- A. $E(Y_t) = \mu$
- B. $E(Y_t) = \mu + \beta_0$
- C. $E(Y_t) = \mu - \beta_1$
- D. $E(Y_t) = \mu - \beta_0$
- E. $E(Y_t) = \mu + \beta_1$

[2]

Q. 10) An insurance distributor which sells life insurance policies for BUI Life insurance company is entitled to 5% commission on renewal premiums. BUI Life has decided to implement a new commission structure based on “conservation ratio” i.e. ratio of current year renewal premiums to total premiums collected in previous year. Under the new structure –

1. If the distributor is able to achieve a yearly conservation ratio of greater than 75%, then he will get an increased commission of 2% on renewal premiums.
2. If he achieves a yearly conservation ratio from 50% to 75%, then his renewal commission will remain at 5% of renewal premiums.
3. If he fails to achieve a yearly conservation ratio of at least 50%, then his renewal commission will be reduced by 2%.
4. Under this structure, minimum commission payable is 3% of renewal premiums and maximum commission payable is 7% of renewal premiums.

If probabilities of achieving a conservation ratio of $<50\%$, between $50\%-75\%$ and $>75\%$ are a_1, a_2 and a_3 respectively, identify the correct one-year transition matrix (for states – 3% commission, 5% commission, 7% commission) for this new commission structure –

A. $\begin{bmatrix} a_1 + a_2 & a_3 & 0 \\ a_1 & a_2 & a_3 \\ 0 & a_1 & a_2 + a_3 \end{bmatrix}$

B. $\begin{bmatrix} a_1 & a_2 + a_3 & 0 \\ a_1 & a_2 & a_3 \\ 0 & a_1 + a_2 & a_3 \end{bmatrix}$

C. $\begin{bmatrix} a_1 & a_2 & a_3 \\ a_1 & a_2 + a_3 & 0 \\ 0 & a_1 & a_2 + a_3 \end{bmatrix}$

D. $\begin{bmatrix} 0 & a_1 & a_2 + a_3 \\ a_1 & a_2 & a_3 \\ a_1 & a_2 + a_3 & 0 \end{bmatrix}$

E. None of the above

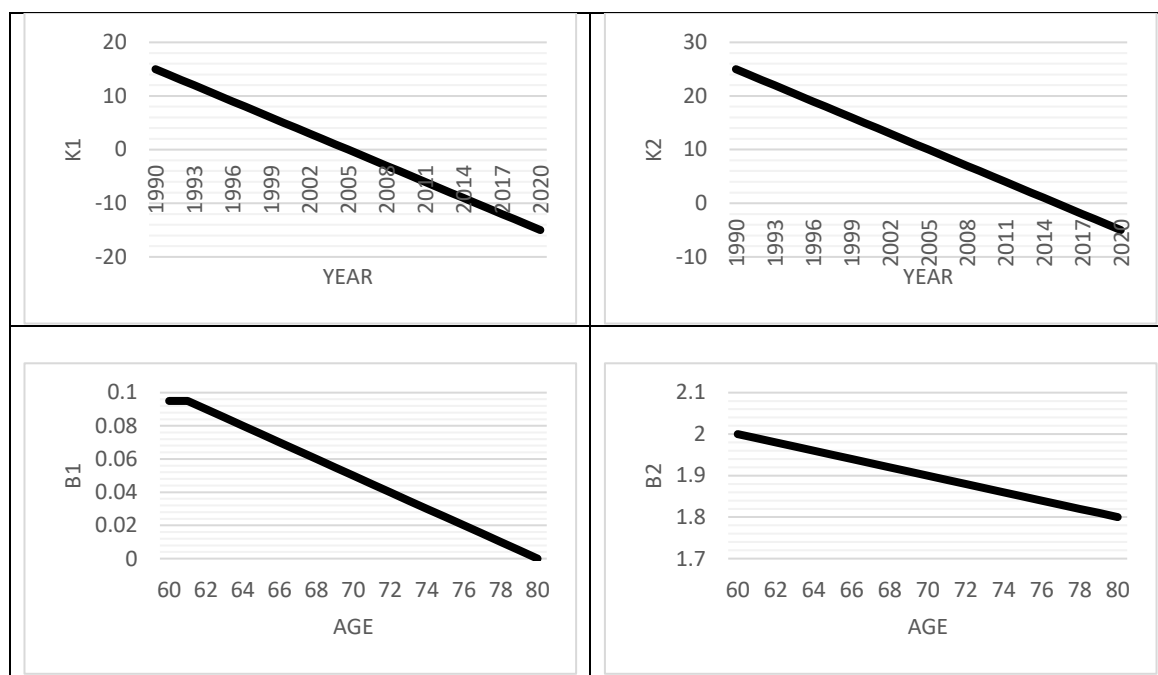
[2]

Q.11) Lee-Carter model is used to model the central mortality rate at age x in year t i.e. $m_{x,t}$.

As per the model, $m_{x,t}$ is dependent on three parameters –

1. a_x describes the general shape of mortality at age x
2. b_x measures the change in the rates in response to an underlying time trend in the level of mortality k_t
3. k_t reflects the effect of the time trend on mortality at time t .

A student actuary has constructed two graphs each for the parameters “ b_x ” and “ k_t ” based on the mortality data from 1990 to 2020 for ages 60 to 80.



By keeping in mind the constraints to be imposed on the Lee-Carter model, which of these graphs are correct representations of the parameters “ b_x ” and “ k_t ”?

- A. $K1$ is a correct representation of k_t and $B1$ is a correct representation of b_x
- B. $K2$ is a correct representation of k_t and $B1$ is a correct representation of b_x
- C. $K1$ is a correct representation of k_t and $B2$ is a correct representation of b_x
- D. $K2$ is a correct representation of k_t and $B2$ is a correct representation of b_x
- E. All of the above.

[2]

Q. 12) Consider the following discrete time stochastic processes defined as follows:

$X_t = X_{t-1} + I_t$	$Y_t = Y_{t-1} + J_t$
$X_0 = 0$	$Y_0 = 0$
And for $t = 1, 2, 3, \dots$	And for $t = 1, 2, 3, \dots$
I_t are random variables taking the values: +1 with probability $p_t = \exp(- X_{t-1})$ and -1 with probability $q_t = 1 - \exp(- X_{t-1})$.	J_t are random variables taking the values: +1 with probability $p = 0.5$ and -1 with probability $q = 0.5$.

Consider the following statements about the two processes:

1. X_t has independent increments.
2. Y_t has independent increments.

3. X_t exhibits the Markov property.
4. Y_t exhibits the Markov property.

Which of the given statements are TRUE?

- A. Statement 1 and 3 only
- B. Statement 1 and 2 only
- C. Statement 3 and 4 only
- D. Statement 2, 3 and 4 only
- E. Statement 1, 3 and 4 only

[2]

Q. 13) Consider the following penalised spline model for projecting mortality rates:

$$\ln(m_{x,t}) = \sum_{j=1}^s \theta_j \beta_j(x) + \lambda * \sum_{j=2}^s (\theta_j - \theta_{j-1})$$

where:

- θ_j are the parameters to be estimated
- $\beta_j(x)$ are the set of splines
- s is the number of splines
- λ is the smoothing parameter
- $\sum_{j=2}^s (\theta_j - \theta_{j-1})$ represents the roughness penalty

If $\lambda = 0$, then which of the following statements is TRUE regarding mortality projections using the penalised spline model?

- A. It will increase the roughness penalty in the mortality curve.
- B. It will increase the roughness in the mortality curve.
- C. It will convert the mortality curve into a straight line.
- D. It will lead to smoother progression in the mortality curve.
- E. It will lead to fitting which closely resembles Gompertz model

[2]

Q. 14) A class of students in high school needs to be divided into two batches – “Elite” and “Zenith” in the year 2025.

“Elite” batch represents those students who are high-performers and need to be given more challenges for their overall development.

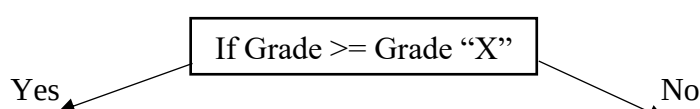
“Zenith” Batch represents average performers who need to be given extra training in order to ensure that they do well.

A binary decision tree algorithm is to be used to do this classification. For this purpose, a training data set of 100 students from the last year is chosen.

The annual assessment data of 100 students for the last year is given below:

Grade	O	A+	A	B+	B	< B
No of students	5	10	15	20	25	25

Following algorithm has been suggested –



Elite Batch

Zenith Batch

Here, Grade “X” can take the values O, A+, A, B, B+.

The school statistics teacher believes that the distribution between the two batches should be equitable and hence the threshold i.e. Grade “X” should be selected in such a way that the Gini Index is minimum.

Based on the training data set, what should be the value of Grade “X” which will ensure minimum Gini Index for this decision tree?

- A. Grade B
- B. Grade B+
- C. Grade A
- D. Grade A+
- E. Grade O

[2]

- Q. 15)** A life insurance company is carrying out mortality investigation of its term assurance portfolio. It records in-force policies using age label “Age last birthday as at 1st April”.

Following information is available for the number of in-force policies:

Date	Age		
	45	46	47
As at 1 st April 2022	18,500	20,000	15,000
As at 1 st April 2023	20,000	21,100	20,700
As at 1 st April 2024	20,100	20,000	19,700

The number of deaths is reported by the claims department and they are calculated based on age x nearest birthday on the date of death.

You may assume that birthdays are uniformly distributed throughout the calendar year.

What will be the force of mortality for lives with age nearest birthday 46, if the number of deaths reported by the claims team for age 46 (nearest birthday) from 1st April 2022 to 1st April 2024 is 2,760?

- A. 6.72%
- B. 6.97%
- C. 6.82%
- D. 7.14%
- E. 7.49%

[2]

- Q. 16)** A product copula for two independent exponentially distributed random variables X and Y with parameters ‘ λ ’ and ‘ μ ’ respectively is denoted by C (u, v). For this copula, following information is obtained regarding the values of X and Y corresponding to the 25th and 75th percentiles for this copula:

n th Percentile	Value of X	Value of Y
25 th Percentile	2.95	1.31
75 th Percentile	5.95	7.00

What are the “method of percentiles” estimates of the parameters λ and μ of the underlying independently distributed exponential distributions?

- A. $\hat{\lambda}_{MOP} = 0.25$ and $\hat{\mu}_{MOP} = 0.75$
- B. $\hat{\lambda}_{MOP} = 0.50$ and $\hat{\mu}_{MOP} = 0.75$
- C. $\hat{\lambda}_{MOP} = 0.50$ and $\hat{\mu}_{MOP} = 0.25$
- D. $\hat{\lambda}_{MOP} = 0.25$ and $\hat{\mu}_{MOP} = 0.50$
- E. $\hat{\lambda}_{MOP} = 0.50$ and $\hat{\mu}_{MOP} = 0.50$

[3]

- Q. 17)** A life insurance company has newly entered into an excess of loss treaty with Actuarial Re with a retention limit of INR 39 lakhs.

Actuarial Re considers that the claims (including the claims directly settled by the life insurer) are independent and have a two-parameter Pareto distribution with parameters α and λ .

The actual amounts settled by Actuarial Re in respect of the life insurer have a mean of INR 6 lakh and a standard deviation of INR 7 lakhs.

What is the proportion of the claims directly settled by the life insurer?

- A. 34%
- B. 42%
- C. 23%
- D. 41%
- E. 29%

[4]

- Q. 18)** The number of years pacemaker batteries can last without replacement can be modelled either using –

1. Exponential distribution with mean of μ .
2. Weibull distribution with parameters c and γ .

Consider the following statements:

1. If $\gamma = 1$ and $c=1/\mu$, the mean failure time in years under both the distributions would be equal.
2. If $\gamma = 0$ and $c=1/\mu$, probability that the batteries fail before time X will be same under both the distributions.
3. If $\gamma > 1$ and $c=1/\mu$, probability that the batteries fail before time X will be higher under the Weibull distribution.
4. If $\gamma < 0$ and $c=1/\mu$, probability that the batteries fail before time X will be lower under the Exponential distribution.

Which of the given statements are TRUE?

- A. Statement (4) only
- B. Statement (1) only
- C. Statement (3) only
- D. Statement (2) only
- E. Statement (1) and (3) only

[2]

- Q. 19)** The market prices of two listed stocks X and Y are denoted by time series processes X_t and Y_t . Both processes are $I(1)$ processes which satisfy the following relations –

$$X_t = aX_{t-1} + bY_{t-1} + e_t^x$$

$$Y_t = bX_{t-1} + aY_{t-1} + e_t^y$$

In the given relations, a and b are non-zero constants and e_t^x and e_t^y are white noise processes.

In which of the following circumstances can X_t and Y_t considered to be cointegrated? Use the cointegrating vector $(1, -1)$.

- A. $a - b < 1$
- B. $a - b > 1$
- C. $b - a > -1$
- D. $b - a < -1$
- E. Both (A) and (C)

[2]

Q. 20) A Space Station “Act-Space” is missioned to launch a satellite capsule which can enter the orbits of either of the two planets “Act-Math” or “Act-Stat”.

Once a satellite capsule is launched at a time point –

- It can enter the orbit of Planet “Act-Math” with a probability of 0.25.
- It can enter the orbit of Planet “Act-Stat” with a probability of 0.25.
- If it misses entering the orbits of either of the planets, it comes back to the space station “Act-Space” with a probability of 0.50. It can be then re-launched at the next time point.

Once, the satellite capsule has entered the orbits of either “Act-Math” or “Act-Stat”, it stays there and cannot come back due to the gravitational pull of these planets.

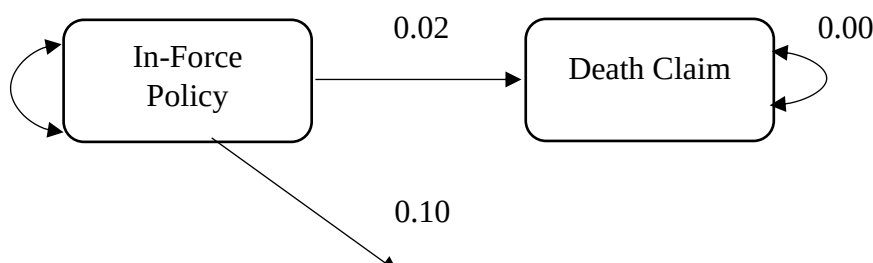
What is the stationary distribution corresponding to the above Markov Chain?

- A. $\pi_{\text{SPACE}} = 0, \pi_{\text{MATH}} = 0.5, \pi_{\text{STAT}} = 0.5$
- B. $\pi_{\text{SPACE}} = 0, \pi_{\text{MATH}} = 0.3, \pi_{\text{STAT}} = 0.7$
- C. $\pi_{\text{SPACE}} = 0, \pi_{\text{MATH}} = 0.6, \pi_{\text{STAT}} = 0.4$
- D. $\pi_{\text{SPACE}} = 0, \pi_{\text{MATH}} = 0.2, \pi_{\text{STAT}} = 0.8$
- E. All of the above

[2]

Q. 21) A life insurance company provides a term assurance cover for a fixed sum assured of INR 1 crore. Regular premium is required to be paid uniformly during the year in respect of the insurance policy. If premium is not paid, the policy lapses and the cover ceases. There is no scope for reviving a lapsed policy.

Claims are modelled using the following model with the one-year transition rates from the various states:





What is the expected cost of claims incurred in the first year in respect of this term assurance policy?

- A. INR 3,20,659
- B. INR 5,47,932
- C. INR 1,88,466
- D. INR 1,80,000
- E. INR 2,00,000

[4]

- Q. 22)** City Care Hospital is pioneering a new surgery to replace human heart with synthetic implants till the time a patient gets a donor. Once the patient gets a donor post the operation, the synthetic implant is replaced with the donor's heart. Operations were carried out throughout March 2025. Patients who underwent the surgery were monitored till the end of April 2025 or till they died or till they got a donor and were operated for replacing the implant with the donor's heart. The results are shown below. Where no date is given, the patient was alive and still in the hospital at the end of April 2025.

Patient	Date of Surgery	Date of leaving	Reason for leaving
Andy	1 st March, 2025	15 th April, 2025	Got a donor
Bernard	5 th March, 2025		
Cherry	15 th March, 2025	27 th March, 2025	Death
Donald	20 th March, 2025	25 th March, 2025	Death
Emily	22 nd March, 2025		
Fred	24 th March, 2025	23 rd April, 2025	Got a donor
George	27 th March, 2025		
Henry	29 th March, 2025	22 nd April, 2025	Death

The management of the hospital wants to know the efficacy of the synthetic implants and hence want you to develop a 90% confidence interval of the probability of surviving for 3 weeks after the implant surgery is done. You are required to use Greenwood's formula for determining the variance of the Kaplan-Meier estimator. Required confidence interval is –

- A. (0.5080, 0.9563)
- B. (0.4863, 0.9780)
- C. (0.4781, 0.9862)
- D. (0.5827, 0.8816)
- E. (0.6126, 0.8517)

[4]

- Q. 23)** A credit risk manager of a non-banking financial company is doing analysis of credit defaults of a few of its loan portfolios. He is trying to identify the types of censoring involved in the investigation viz. Type I censoring, Type II censoring, Informative censoring or Non-Informative censoring.

Consider the following details regarding the analysis:

1. The study covers a period of 5 years from 2020 to 2024 and ends in 2024 regardless of whether the firm being studied has defaulted or not.

2. As per the NBFC's articles of association, for a particular loan portfolio, if there are defaults equal to more than 20% of the value of the portfolio, then the entire portfolio has to be liquidated and hence the analysis has to be stopped in respect of that portfolio.
3. If the CIBIL score of the individual borrower from a particular loan portfolio exceeds 750, then the individual is not considered for the purpose of credit analysis.
4. The analysis specifically excludes the pandemic period from April 2020 to September 2021 for all borrowers from all portfolios under study.

Which of the following pairs are correct?

	Type 1 censoring	Type II censoring	Informative censoring	Non-Informative censoring
A.	1	2	3	4
B.	2	3	1	4
C.	4	1	2	3
D.	1	4	3	2
E.	2	1	3	4

[2]

Q. 24) GETFLIX, a popular OTT platform is doing subscription retention study based on 2024 data. It is studying the rate at which subscribers unsubscribe the platform during the year by fitting a Cox proportional hazards model. In the fitted model, the hazard depends on the following three covariates –

- Proportion of GETFLIX originals during the year: $<40\% = 0$, $\geq 40\% = 1$.
- Social media buzz: Against = 0, For = 1.
- Subscription cost: $> 120\%$ of the cheapest OTT platform = 1, $\leq 120\%$ of the cheapest OTT platform = 0.

In 2023, using the same three parameters (and the same criteria), the subscriber retention rate was calculated as 65% using the fitted model.

During 2023 –

- Of the total streaming hours available 50% were GETFLIX originals.
- A popular social media influencer had signed promotions related agreements with GETFLIX promoting in the favour of GETFLIX content.
- GETFLIX was the market leader in terms of subscription cost.

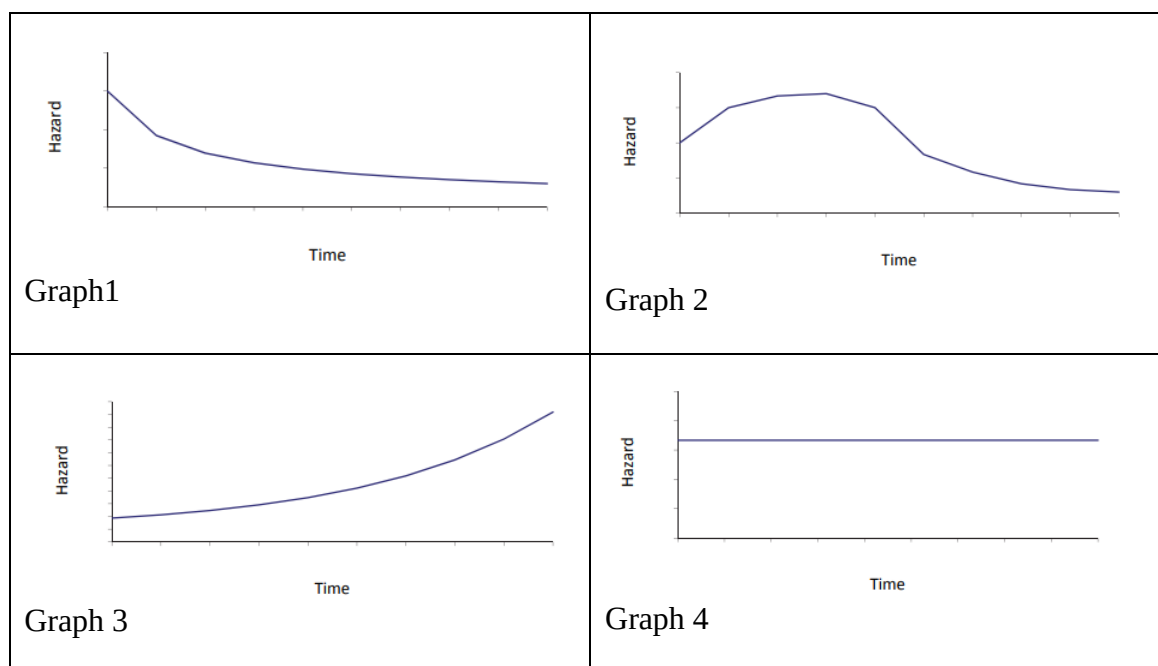
During the year 2024, GETFLIX has increased its fees by 30% of its previous year's fees and the nearest cost competitor last year has become the new market leader. It was charging 5% more than GETFLIX in 2023 and has reduced its fees by 20% of its previous year's fees.

If there is no change in the first two parameters and their coefficients and if the coefficient for the subscription cost parameter is -0.25, predict the subscriber retention rate for GETFLIX for the year 2024:

- A. 28%
- B. 44%
- C. 46%
- D. 72%
- E. 57%

[3]

Q. 25) Following are the graphs of hazard functions, with hazard plotted on Y-axis and time plotted on X-axis:



Which of the following pairs are correct?

	Constant Hazard	Monotonically Decreasing Hazard	Exponential Hazard	Log-logistic Hazard
A.	Graph 1	Graph 2	Graph 3	Graph 4
B.	Graph 2	Graph 3	Graph 4	Graph 1
C.	Graph 4	Graph 1	Graph 2	Graph 3
D.	Graph 4	Graph 1	Graph 3	Graph 2
E.	Graph 3	Graph 1	Graph 4	Graph 2

[2]

Q. 26) A three-year critical illness policy sold by a standalone health insurance company is modelled using the following assumptions:

- Number of claims under the policy during the term are equally likely to be 0, 1 or 2.
- Individual claim amounts are equally likely to be INR 1 lakh or INR 2 lakh.
- The company has a premium loading of 20%.

The health insurance company is thinking of introducing a policy excess of INR “E” per claim in return for a reduction in policy premium by INR 10,000.

What is the minimum value of “E” for which the insurance company would be better off as compared to the existing scenario (scenario without policy excess)?

- A.** INR 15,000
- B.** INR 30,000
- C.** INR 20,000
- D.** INR 25,000
- E.** INR 10,000

[2]

Q. 27) A non-life insurance company has entered into a proportional reinsurance arrangement with a reinsurer Statistica Re. in respect of its home contents insurance policy. Following are the details of the arrangement –

- The direct insurer retains first claim under the policy and from the second claim, the claims are passed on to Statistica Re.
- Statistica Re. covers only 50% of the claim amount from the second claim and the balance 50% is retained by the direct insurer.

The number of claims N follow a type 2 geometric distribution with parameter 0.50 so that:

$$P(N = n) = 0.50 * 0.50^n \quad n = 0, 1, 2, \dots$$

Individual claim amounts have a mean of INR 5,000.

Let S_R denote the aggregate claim to be settled by the reinsurer. What is the per policy expected reinsurance claim pay-out i.e. $E(S_R)$ for Statistica Re.?

- A. INR 1,250
- B. INR 1,875
- C. INR 625
- D. INR 2,500
- E. INR 750

[4]

Q. 28) A student actuary who wants to model catastrophic losses has chosen exponential distribution for modelling tail events. He has calculated measures of tail weight relating to an exponential distribution and has made the following assertions:

1. The hazard rate for an exponential distribution is increasing and hence it corresponds to a lighter tail.
2. The mean residual life for an exponential distribution is decreasing and hence it corresponds to a lighter tail.
3. Existence of moments $E(X^k)$ for all positive integers k indicates that the exponential distribution has a heavy tail.
4. The mean residual life is the multiplicative inverse of hazard rate in case of an exponential distribution.
5. An exponential distribution with parameter $\lambda = 2$ has a much thicker tail as compared to an exponential distribution with parameter $\lambda = 3$ as the limiting density ratio of these two distributions tends to zero.

Which of the given statements is correct?

- A. Statement 1
- B. Statement 2
- C. Statement 3
- D. Statement 4
- E. Statement 5

[3]

Q. 29)

The state of financial market in a country is modelled using a time in-homogenous Markov jump process with state space = $\{1 = \text{bull market}, 0 = \text{bear market}\}$. The transition rate for switching between the states is a function of time t but it is independent of the state which

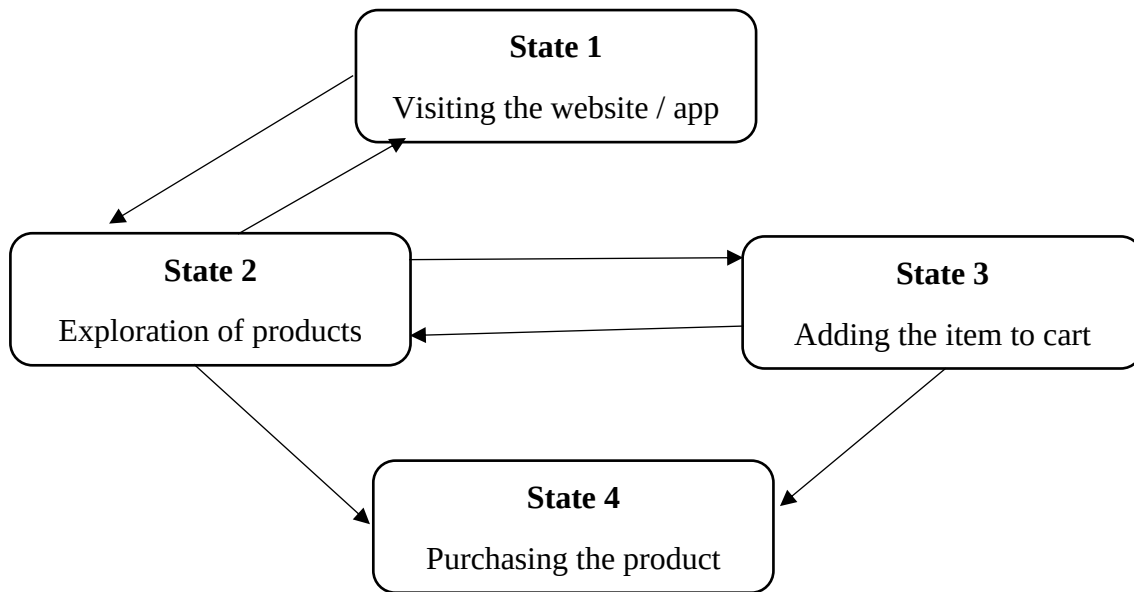
is currently occupied by the process. The process starts in state 0 (bear market) at time 't' = 0. The probability that the process remains in bear market state till time 's' is $\exp(-s^2)$.

What is the transition rate of this process?

- A. t
- B. 2t
- C. t^2
- D. t+1
- E. 1

[2]

Q. 30) An e-commerce portal Clipkart wants to model a user's purchase journey using a four-state time in-homogenous Markov jump process represented by the following graph:



The force of transition from state i to stage j ($i \neq j$) at age x is denoted by μ_x^{ij} and the probability that a user who is in state i at time x , will be in state j at time $(x + t)$ is ${}_t p_x^{ij}$.

Which of the following represents a correct expression of the Kolmogorov's forward differential equation for ${}_t p_x^{23}$?

- A. $d/dt ({}_t p_x^{23}) = {}_t p_x^{21} \mu_{x+t}^{13} + p_x^{22} \mu_{x+t}^{23} - p_x^{23} \mu_{x+t}^{32}$
- B. $d/dt ({}_t p_x^{23}) = {}_t p_x^{21} \mu_{x+t}^{13} - p_x^{22} \mu_{x+t}^{23} - p_x^{23} \mu_{x+t}^{32}$
- C. $d/dt ({}_t p_x^{23}) = {}_t p_x^{22} \mu_{x+t}^{23} - p_x^{23} \mu_{x+t}^{34} - p_x^{23} \mu_{x+t}^{32}$
- D. $d/dt ({}_t p_x^{23}) = {}_t p_x^{21} \mu_{x+t}^{13} + p_x^{22} \mu_{x+t}^{23} - p_x^{23} \mu_{x+t}^{34}$
- E. $d/dt ({}_t p_x^{23}) = {}_t p_x^{22} \mu_{x+t}^{23} + p_x^{23} \mu_{x+t}^{34} + p_x^{23} \mu_{x+t}^{32}$

[3]

Q. 31) A time series process is used to model the vote share of DJP in a state assembly election based on historical data for the past 20 elections in the state. The process is as follows:

$$(X_t - X_{t-1}) = \alpha (X_{t-1} - X_{t-2}) + e_t$$

In the given process, e_t is a white noise process with mean 0 and variance equal to σ^2 . Also, e_t is independent of X_t .

You are given the following aggregates based on the historical data:

$$\sum_{i=2}^{20} (x_i - x_{t-1})^2 = 4, \sum_{i=2}^{20} (x_i - x_{t-1}) (x_{t+1} - x_t) = 3$$

The vote share in the latest two elections is: $x_{20} = 39.2\%$ and $x_{19} = 42.1\%$.

Using the Box-Jenkins methodology, the forecasted value of the vote share of DJP in the next (21st) state assembly election is –

- A. 37.025%
- B. 41.375%
- C. 43.067%
- D. 35.333%
- E. None of the above

[2]

Q. 32) An actuary wants to graduate the mortality rates for a term assurance portfolio of a life insurance company for ages from 21 to 50. The actuary is doing graduation by referring to standard mortality table IALM (Indian Assured Lives Mortality) 2012-14.

It is decided to fit the following link functions between for graduating mortality rates:

$$\mu_x^G = \mu_{x-k}^S \quad \text{for } k = 1, 2, 3, 4, 5.$$

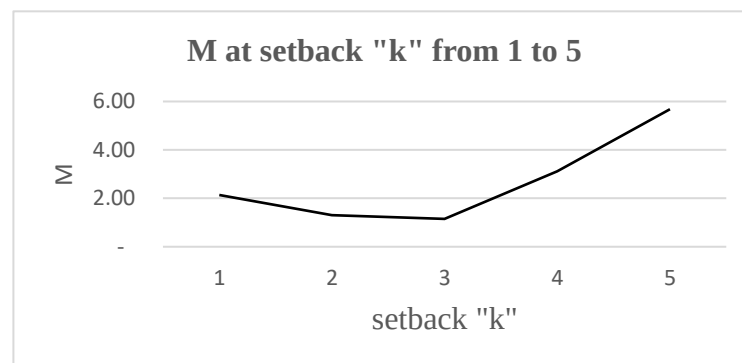
Where:

- μ_x^G = graduated mortality rate for a life aged x
- μ_x^S = mortality rate for a life aged x as per IALM 2012-14.
- k = the age setback
- $\hat{\mu}_x$ = calculated mortality rate for a life aged x based on the company's experience

In order to determine a better fit, the actuary has determined the statistic:

$$M = \sum \text{var}(\hat{\mu}_x) * (\mu_x^G - \hat{\mu}_x)^2.$$

M has been plotted on the y-axis and the set-back has been plotted on x-axis.



What level of age setback gives a better fit?

- A. Setback of 1 year
- B. Setback of 2 years
- C. Setback of 3 years
- D. Setback of 4 years
- E. Setback of 5 years

[2]

Q. 33) Health-Fit is a health awareness platform and runs an application “H-FIT” to monitor exercise, calorie intake, rest and recovery. A person suffering from uncontrolled obesity has recently installed the application.

The person was surprised to see the results:

- In the first 30 days, the person lost 20% of his initial weight
- In the next 60 days, the person lost 20% of his weight on the 31st day.

Assuming that the force of weight loss “ μ_1 ” and “ μ_2 ” for the first 30 days and the next 60 days respectively are constant across the respective periods, determine the value of “ μ_1 ” and “ μ_2 ”.

	μ_1 for first 30 days (per 1,000)	μ_2 for next 60 days (per 1,000)
A.	6.26	3.13
B.	3.72	3.72
C.	7.44	7.44
D.	7.44	3.72
E.	3.13	6.26

[2]

- Q. 34)** The number of phishing emails that an organization receives per day is modelled as a time series process as follows:

$$Y_t = \alpha * Y_{t-1} + Z_t$$

In the given process, Z_t is a zero mean white noise process with a variance of σ^2 .

What is the probability that on any given day, the number of phishing emails received by the organization is greater than 10 i.e. $P(Y_t > 10)$? You may assume $\alpha = 0.50$ and $\sigma^2 = 48$.

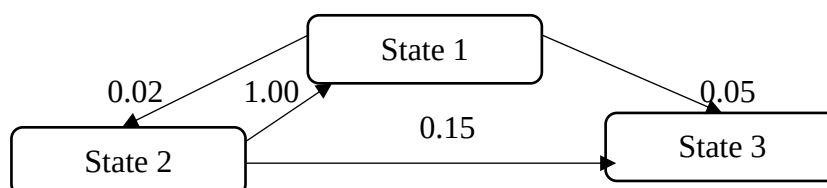
- A. 3.216%
- B. 7.533%
- C. 11.507%
- D. 27.425%
- E. None of the above

[3]

- Q. 35)** An air-conditioner supplier has developed a three state Markov model relating to its central air-conditioner units. The three states are as follows:

- State 1: when the air-conditioner unit is functioning properly.
- State 2: when the air conditioner unit has some defects and hence require repairs.
- State 3: when the air conditioner unit is totally dysfunctional.

Following is the graph along with transition rates per year:



What is the expected lifetime (in years) of one air conditioner unit sold by the supplier?

- A. 25 years

- B. 19 years
- C. 15 years
- D. 12 years
- E. 10 years

[3]

Q. 36) Rainfall received per day (in centimetres) by a town is modelled using a Burr distribution with parameters $\alpha = 2$, $\lambda = 2$, $\gamma = 2$. What is the variance of daily rainfall?

- A. $4 - 0.125 \pi^2$
- B. $2 - 0.125 \pi^2$
- C. $2 - 0.125 \pi$
- D. $4 - 0.125 \pi$
- E. $4 - 0.625 \pi^2$

[2]

Q. 37) Support requests received at the information technology support centre of a public sector bank are of two types –

Type 1: Request for inability to connect to the bank's network

Type 2: Requests for resetting of passwords (on expiry of old password).

Both types of requests follow a Poisson process with rates of 50 per day and 40 per day respectively. The Poisson processes for the two types of requests are assumed to be independent of each other.

For a randomly selected Type 1 request, the probability of the response time being more than a minute is $1/10$ while for a randomly selected Type 2 request, the probability of the response time being more than a minute is $1/20$.

On a given day, how many requests received by the support centre will require a response time of more than a minute?

- A. 1
- B. 2
- C. 5
- D. 7
- E. 10

[2]

Q. 38) Probability that an individual aged 35 years will survive for the next 10 years using Makeham's law (${}_{10}p^M_{35}$) is 75% of the probability that the individual will survive for the next 10 years using Gompertz' law (${}_{10}p^G_{35}$).

After 10 years, when the individual is aged 45, which of the following is TRUE regarding the probability that the same individual survives for 10 more years?

- A. ${}_{10}p^M_{45} = 75\% * {}_{10}p^G_{45}$
- B. ${}_{10}p^M_{45} > 75\% * {}_{10}p^G_{45}$
- C. ${}_{10}p^M_{45} < 75\% * {}_{10}p^G_{45}$
- D. ${}_{10}p^M_{45} = 75\% * {}_{10}p^G_{35}$
- E. ${}_{10}p^M_{35} = 75\% * {}_{10}p^G_{45}$

[2]

Q. 39) Under a two-state Poisson model of mortality, the number of deaths D has a Poisson distribution with parameter equal to $(\mu * E^C_x)$ where μ is the force of mortality and E^C_x is the population exposed to mortality risk.

The maximum likelihood estimator $\hat{\mu}_{MLE} = D / E_x^C$ is obtained in such a way that it maximizes the log likelihood function: $\log L(\mu) = d(\log \mu + \log E_x^C) - \mu E_x^C - \log d!$

What is the Cramer-Rao's lower bound for the variance of the estimator $\hat{\mu}_{MLE}$?

- A. $CRLB(\hat{\mu}_{MLE}) = D / E_x^C$
- B. $CRLB(\hat{\mu}_{MLE}) = D^2 / E_x^C$
- C. $CRLB(\hat{\mu}_{MLE}) = (D / E_x^C)^2$
- D. $CRLB(\hat{\mu}_{MLE}) = (\mu / E_x^C)^2$
- E. $CRLB(\hat{\mu}_{MLE}) = \mu / E_x^C$

[2]

Q. 40) Following pairs have been incorrectly matched:

	Copula		Key Feature
A1	Gumbell copula	B1	A high degree of positive interdependence throughout.
A2	Student's t copula	B2	High lower tail dependence, no upper tail dependence.
A3	Co-monotonic copula	B3	Both upper and lower tail dependence.
A4	Clayton copula	B4	High upper tail dependence, no lower tail dependence.

Which of the following represents correctly matched pairs?

A.	A1-B2;	A2-B1;	A3-B4;	A4-B3.
B.	A1-B2;	A2-B3;	A3-B1;	A4-B4.
C.	A1-B4;	A2-B3;	A3-B1;	A4-B2.
D.	A1-B3;	A2-B4;	A3-B2;	A4-B1.
E.	A1-B4;	A2-B2;	A3-B3;	A4-B1.

[2]

Q. 41)

The office of Economic Advisor of Economa, a developing economy is developing a multi-variate linear regression model for predicting the consumption expenditure (Y) during the year based on two explanatory variables viz. national income (X_1) and government spending (X_2).

The variables X_1 and X_2 have been standardized by subtracting the sample means from each value and dividing it by the sample standard deviation.

The model is assumed to pass through the origin and hence the intercept parameter β_0 can be assumed to be equal to zero.

The ordinary least square estimates of the regression coefficients β_1 and β_2 have been calculated as 4 and 3 respectively by minimizing the total sum of squares (TSS) which is given by the following expression:

$$TSS = \sum (Y - (\beta_1 X_1 + \beta_2 X_2))^2$$

Considering that national income and government spending are correlated, to reduce the multi-collinearity of the model and to address the problem of overfitting, ridge regression

technique is proposed to be applied and the new regression coefficients β_{R1} and β_{R2} are to be arrived at by minimizing the following revised TSS expression:

$$\text{TSS}_R = \sum (Y - (\beta_{R1}X_1 + \beta_{R2}X_2))^2 + \lambda * (\beta_{R1}^2 + \beta_{R2}^2)$$

If $\sum X_1X_2 = 20$, $\sum X_1^2 = 10$, $\sum X_2^2 = 12$ and $\lambda = 5$, what will be the values of the new regression coefficients β_{R1} and β_{R2} ?

- A. $\beta_{R1} = 3.20$ and $\beta_{R2} = 2.60$
- B. $\beta_{R1} = 4.60$ and $\beta_{R2} = 3.30$
- C. $\beta_{R1} = 2.90$ and $\beta_{R2} = 2.80$
- D. $\beta_{R1} = 1.20$ and $\beta_{R2} = 4.30$
- E. $\beta_{R1} = 1.80$ and $\beta_{R2} = 3.20$

[3]
